

Scientific American Article

Writing Prompt

(3+:0)

More than three decades of research have shown that a focus on effort – not intelligence or ability – is key to success in school and life. Read carefully “The Secret to Raising Smart Kids.” Then, in a one-chunk paragraph (3+:0), summarize the “Two Views of Intelligence.” *For your concluding sentence, explain how this information might affect a student’s life.*



YOUR PRACTICE



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The Secret to Raising Smart Kids

Hint: don't tell your kids that they are. More than three decades of research shows that a focus on effort - not on intelligence or ability - is key to success in school and in life.

By Carol S. Dweck

- [1] Our society worships talent, and many people assume that possessing superior intelligence or ability - along with confidence in that ability - is the recipe for success. In fact, however, more than 30 years of scientific investigation suggest that an over emphasis on intellect or talent leaves people **vulnerable** to failure, fearful of challenges and unwilling to remedy their shortcomings.
- [2] The result plays out in children who coast through the early grades under the dangerous notion that no-effort academic achievement defines them as smart or gifted. Such children hold an **implicit** belief that intelligence is **innate** and fixed, making striving to learn seem far less important than being (or looking) smart. This belief also makes them see challenges, mistakes and even the need to exert effort as threats to their **ego** rather than as opportunities to improve. And it causes them to lose confidence and motivation when the work is no longer easy for them.
- [3] Praising children's innate abilities, reinforces this mindset, which can also prevent young athletes or people in the workforce and even marriages from living up to their potential. On the other hand, our studies show that teaching people to have a "growth mindset," which encourages a focus on effort rather than on intelligence or talent, helps make them into high achievers in school and in life.

The Opportunity of Defeat

- [4] I first began to investigate the underpinnings of human motivation - and how people **persevere** after setbacks - as a **psychology** graduate student at Yale University in the 1960s. Animal experiments by psychologist Martin Seligman, Steven Majer and Richard Solomon of the University of Pennsylvania had shown that after repeated failures, most animals conclude that the situation is hopeless and beyond their control. After such an experience, the researchers found, an animal often remains **passive** even when it can affect change - a state they called learned helplessness.

Vocabulary

vulnerable - open to moral attack, criticism, temptation, etc.

implicit - unquestioning or unreserved; absolute.

innate - existing in one from birth; inborn; native.

ego - self esteem or self image; feelings.

persevere - to persist in anything undertaken; maintain a purpose in spite of difficulty, obstacles or discouragement; continue steadfastly.

psychology - the science of human and animal behavior.

NOTES

NOTES

- [5] *People* can learn to be helpless, too, but not everyone reacts to setbacks this way. I wondered: Why do some students give up when they encounter difficulty, whereas others who are no more skilled continue to strive and learn? One answer, I soon discovered, lay in people's beliefs about why they had failed.
- [6] **Two Views of Intelligence**
Several years later I developed a broader theory of what separates the two general classes of learners - helpless versus mastery-oriented. I realized that these different types of students not only explain their failures differently, but they also hold different "theories" of intelligence. The helpless ones believe that intelligence is a fixed trait: you have only a certain amount, and that's that. I call this "fixed mind-set". Mistakes crack their self-confidence because they attribute errors to a lack of ability, which they feel powerless to change. They avoid challenges because challenges make mistakes more likely and looking smart less so. Such children shun effort in the belief that having to work hard means they are dumb.
- [7] The mastery-**oriented** children, on the other hand, think intelligence is **malleable** and can be developed through education and hard work. They want to learn above all else. After all, if you believe that you can expand your intellectual skills, you want to do just that. Because slip ups stem from a lack of effort, not ability, they can be remedied by more effort. Challenges are energizing rather than **intimidating**; they offer opportunities to learn. Students with such a growth **mind-set**, we predicted, were destined for greater academic success and were quite likely to outperform their counterparts.
- [8] We **validated** these expectations in a study published in early 2007. Psychologist Lisa Blackwell of Columbia University and Kali H. Trzesniewski of Stanford University and I monitored 373 students for two years during the transition to junior high school, when the work gets more difficult and the grading more **stringent**, to determine how their mindsets might affect their math grades. At the beginning of seventh grade, we assessed students' mind-sets for asking them to agree or disagree with statements such as "Your intelligence is something very basic about you that you can't really change." We then assessed their beliefs about other aspects of learning and looked to see what happened to their grades.
- [9] As we predicted, the students with a growth mindset felt that learning was a more important goal in school than getting good grades. In addition, they held hard work in high regard, believing that the more you labored at something, the better you would become at it. They understood that even geniuses have to work hard for their great accomplishments. Confronted by a setback such as a disappointing test grade, students with a growth mind-set said they would study harder or try a different strategy for mastering the material.

Vocabulary

passive - not reacting visibly to something that might be expected to produce manifestations of an emotion or feeling.

oriented - to make familiar with or adjusted to facts, principals, or a situation.

malleable - adaptable or traceable.

intimidating - to force into or deter from some action by inducing fear.

validated - to give official sanction, confirmation, or approval to.

stringent - rigorously binding or exacting; strict; severe.

assessed - to determine the value, significance, or extent of; appraise.

MAIN IDEA SHEET STEPS

1. Write the title of the article in quotation marks on the space provided. Tell students the title is often a clue to the main idea. Have them make a guess at what the article might be about.
2. Read the article, and underline in black a sentence or phrase or section that sums up the main idea or is the thesis of the article. Quote that sentence/section or paraphrase in the space for the main idea. (Reflect on whether or not their predictions about the main idea were correct. Was the title helpful or was it misleading?) If the article doesn't explicitly state a main idea, then have students find the main ideas of each paragraph or section of the article and jot notes about that next to the paragraphs. Use that information to determine a main idea.
3. Next, underline two CDs in the article that support the main idea and write them in the space provided on the sheet. Be sure to cite the page number or paragraph number for the CD if appropriate.
4. For argumentation or explanatory pieces, write some CM nuggets next to the CDs on the article to start thinking about the importance/significance/implications of the specific evidence you chose. These annotations will help students formulate a thesis statement for their papers.
5. For narrative pieces, use the title to help determine the main idea of the story. Underline two CDs that show the main idea or the main action of the story.
6. For poetry, use the title to help determine the main idea of the poem. Underline two CDs that show the main images or the main action of the poem.

**Finding the Main Idea**

Source #1:	_____
Main Idea:	_____
CD	_____

CD	_____

Source #2:	_____
Main Idea:	_____
CD	_____

CD	_____

Source #3:	_____
Main Idea:	_____
CD	_____

CD	_____

Source #4:	_____
Main Idea:	_____
CD	_____

CD	_____

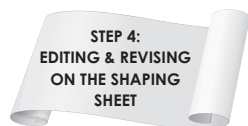


**T-CHART
(3+:0)**

TOPIC SENTENCE: _____

CDs	CMs

CONCLUDING SENTENCE: _____



Name _____

Date _____

SHAPING SHEET

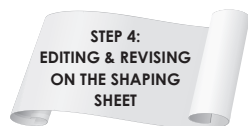
(3+:0)



3+

CDs





Name _____

Date _____

SHAPING SHEET

(3+:0)



A large trapezoidal box with a thick black border. Inside, there are three horizontal lines for writing.

3+

CDs

A large rectangular box with a thick black border. Inside, there are ten horizontal lines for writing.



A large trapezoidal box with a thick black border. Inside, there are three horizontal lines for writing.



PARAGRAPH FORM

[illegible]

FORMAL LAB REPORT FORMAT

The following is a guide for all formal lab reports in this course. You need not limit yourself to this outline. If additional information is warranted, then please add it.

Title

- The title should indicate what the lab is all about. It is usually written in an IV vs. DV format. Centered at the top of the page.

Introduction & Background

- Indicate what you hope to learn (purpose of the lab)
- Clearly identify what you think is going to happen (hypothesis)
- Include important characteristics of the IV and DV (applicable to the study) - with citations

Living Organism

- Scientific Name
- Habitat - could include location in the world and specific ecosystem
- Lifestyle - characteristics about how it lives (nocturnal)
- Anatomy - anything that is applicable to the study
- Physiology - relate to the anatomy
- Feeding - what and how it eats (mode of nutrition)
- Behavior - mating rituals, communication, etc...

Energy

- Definition
- Form
- Produced
- Measured

Matter

- Name
 - Formula
 - Chemical properties
 - Atomic properties
 - Physical properties
 - History
 - Uses
 - Occurrence - where is it found and "life" span
 - Production - how it is produced if not a natural element
- Previous related studies and their findings
 - Caveat: This introduction section will take some research. Do not "wing it" by making up information from your head!!

Materials and Methods

- A brief description of the procedure (major steps) to show how the lab was conducted, including materials used
- List of components of experimental design: independent variable, dependent variable, control factors, constants
- Describe methods for controlling variables
- Describe methods for collecting data
- Note any departure from the instructions given
- Drawing or pictures can be added if set up is unique

Results (present the data)

- Construct all data tables and charts to present data collected
- Titles and labels for all tables and charts must be included
- Construct all graphs needed to show results
 - Correct type (bar, line, pie)
 - Title
 - Appropriate units
 - Labeled axis
 - Legend
- Labeled diagrams or photos

Conclusion & Discussion

- What was the purpose?
- What happened?
- Based on the data, state a valid conclusion and explain why you think the results turned out the way they did.
- Explain any unexpected results and why those results may have been obtained (error analysis).
- How do the results compare to the research?
- Recommendations for further study

FORMAL LAB REPORT FORMATTitle

Introduction & BackgroundWhat do you hope to learn (purpose of the lab)?

What you think is going to happen (hypothesis)?

Include important characteristics of IV and DV (applicable to the study) – with citations.

◦ Living Organism

• Scientific Name

• Feeding Habits

• Lifestyle

• Anatomy

• Physiology

- Habitat _____

- Behavior _____

- Energy
 - Definition _____

 - Form _____

 - Produced _____

 - Measured _____

- Matter
 - Name _____
 - Formula _____



LAB REPORT

JSWP COLOR-CODED GRADING RUBRIC

By Amy Cote, AP Biology Consultant

By using this rubric, you should be able to know what your grade will be for your lab prior to submitting it. Make sure you have all of these items in your report.

REPORT ITEM	POSSIBLE POINTS
Title	3
Introduction	(20)
Purpose of lab	5
Hypothesis	3
Background info	10
Previous related study	2
Materials & Methods	(18)
Materials listed*	5
Clearly stated procedures*	5
Experimental design	4
Controlling variables	4
Results	(24)
Data in tables	10
Graphs present	10
Title/labels	2
Axis labeled correctly	2
Conclusion	(25)
Summary of results	4
Interpretation of results	12
Results compared to hypothesis	2
Errors analysis	2
Recommendations for further study	5
Citations	(4)
Citation within write-up	2
Accuracy of citation information	2
Correct Use of Language	(6)
Grammar	2
Punctuation	2
Spelling	2
TOTAL	100

Lab Write-Up Rubric

Designed and JSWP Color-Coded by Dhahran High School Science Department

NAME:	LAB REPORT SECTION	POINTS POSSIBLE	POINTS EARNED	COMMENTS
PERIOD:	LAB DESIGN			
	Heading: Include: (a) title of the lab (centered); (b) your name, (right); (c) the names of your lab partners (right); and, (d) the date(s) (right) the lab was performed.	.5		
	Research Question: The question should be (a) focused and (b) quantifiable.	1		
Lab Format: Font: 10 pt Times New Roman. 1.5 spaced. 1 inch margins Should be formally written, do not use "I", "You" or "We" NOTES TO STUDENT: Look for bolded Schaffer Writing prompts (i.e. 3+:0, 2+:1, etc.)	Background Information: (a) Connect how this lab report relates to what we're studying in class right now. (2+:1) (b) What information and vocabulary would a reader need to know to understand your report (cite sources)?	1		
	Variables: (a) list independent, dependent (w/ units) and (b) controlled variables.	2		
	Hypothesis: (a) Answer the research question using the independent and dependent variable. (b) Cite the sources that guided you to your answer.	2		
	Labeled Diagram: (a) Sketch and (b) label an appropriate diagram of your lab set up. Show how someone would set up your lab in the same way.	1		
	Materials: (a) What did you use to conduct your experiment? (b) List amounts and numbers for all materials (The materials should all be referenced in your procedure!).	1		
	Procedure: Use a numbered list to describe the steps to performing the experiment. (Be sure you repeat the experiment more than once (multiple trials). It should read like a cookbook. Someone should be able to read it and get the same results you did.)	1		
	Method for Controlling Variables: For each control variable listed, state how you are going to keep it the same from trial to trial.	1		
	DATA COLLECTION AND PROCESSING			
	Data Collection: (a) Put all your raw data neatly in a table; be sure to include all quantitative and qualitative raw data. (Consult guidelines on back!)	1		
	Data Processing: (a) The data will need to be processed in some way, such as averaging, squaring, dividing, etc., in order to prepare it for graphing. (b) Include this information in a second table. (Consult guidelines on back!)	1		
	Graphs: (a) Choose the right graph! Be sure to label the axis and include a title. Units. (b) Don't forget a legend, too! (Consult guidelines on back!)	1		
	CONCLUSION AND EVALUATION			
	Analysis: Please! (a) Restate the question, hypothesis, and paraphrase the procedure. (b) In multiple chunks, discuss how your results differed or supported your hypothesis, citing your data! (1:2+)	2		
	Evaluation: Discuss all relevant sources of error and their effects on your results. (2+:1)	1		
	Improvement: Explain how to modify the experimental technique if needed. (These improvements should specifically relate to the Evaluation section.)	1		
	Conclusion: Based on the evidence stated above, the hypothesis is / is not supported.	2		
	Bibliography: Use Modern Language Association (MLA), the same format you use in English Class.	.5		
		20 POINTS POSSIBLE		<=Your Score

Notes

All Students Can Think; All Students Can Write.

*“Educating the mind without educating the
heart is no education at all.”
Aristotle*